CSE 311:
Data Communication

Instructor:

Dr. Md. Monirul Islam

Course Outline

- Introduction to digital data communication;
- Introduction to signals
- Review of Fourier Transform
- Frequency Response of Linear Systems;
- Sampling theorem; Quantization;
- Line coding;
- Analog Modulation
- Pulse modulation
- Intersymbol interference;
- Pulse shaping;
- Digital modulation
- Multiple access techniques: TDM, FDM; Random processes;
- Additive White Gaussian Noise (AWGN);
- Error rate due to noise
- Introduction to information theory; Concept of channel coding and capacity.

Course Outcome

- have in-depth knowledge and understanding of analog and digital Data Comm. techniques
- identify and compare pros and cons of different CoE techniques
- analyze real world CoE problems and apply appropriate algorithm(s) to formulate solutions
- design and implement core CoE techniques and
- develop/engineer new techniques for solutions of real world problems

Text Books

1. **Modern Digital and Analog Communication Systems**, *International* 4th edition, B P Lathi and Zhi Ding
2. **An Introduction to Analog and Digital Communications**, 2nd edition, S. Haykin and M. Moher

Assessment

- Class Tests: 20%
- Attendance:10 %
- Term final: 70%

Assessment

- 2 Class Tests in the first part
  - 4^{th} week : Sunday
  - 6^{th} week: Sunday

in absence of central CT routine

Motivation

- CoE made life easy

- Most of the PhD’s in the Department from communication background
  - Funding is easier

- See: Maths are working

- Application to other fields
  - Computer networks
  - Digital signal processing
  - Image/video processing
  - Computer vision

Communication systems

- Continued from ancient times
  - Runners
  - Carrier pigeons
  - Lights
  - Smoke and fire

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Communication systems

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  - Runners
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    - Slow
    - Unreliable

      - However, that was enough for the time

Communication systems

- Electrical communication systems
  - Reliable
  - dependable
  - economical

Communication systems

- **Electrical communication systems**
  - **Reliable**
  - **dependable**
  - **economical**

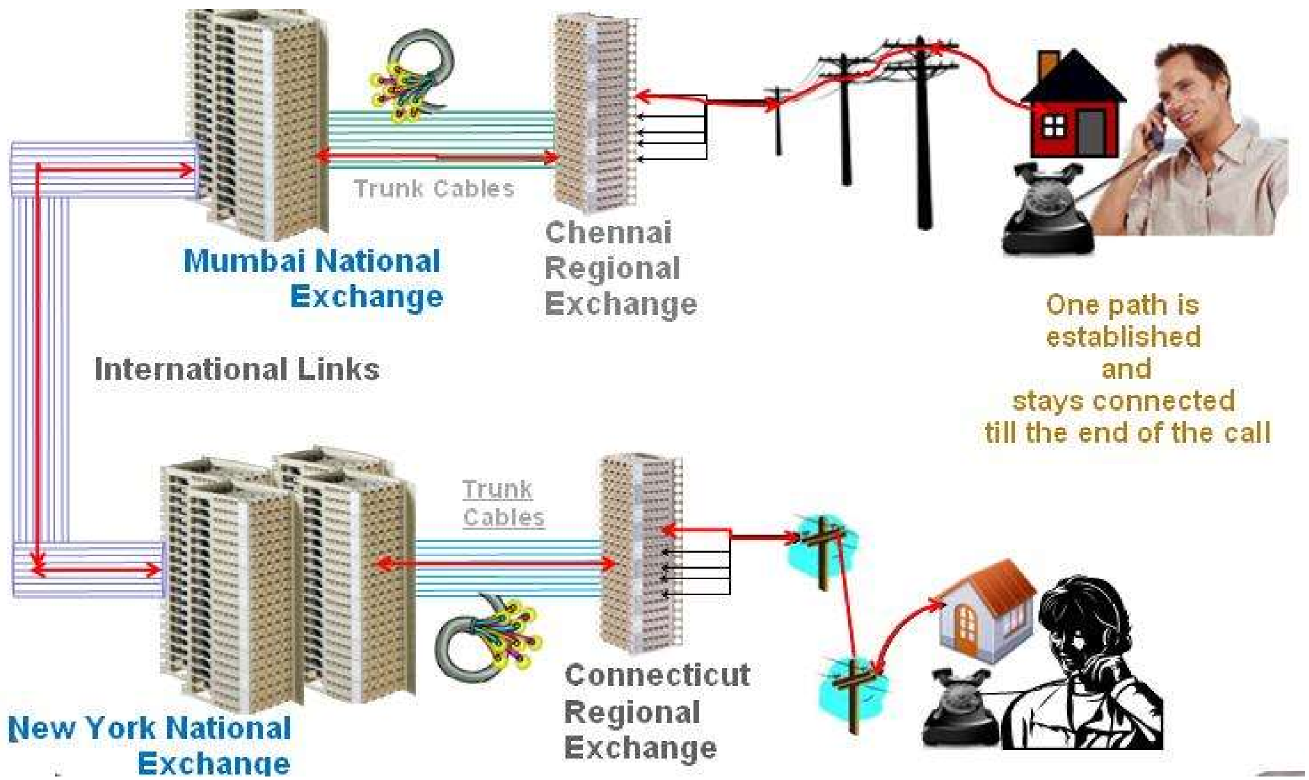
    - **Email**
    - **E-commerce**
    - **Intercontinental business meeting**

Communication system breakthroughs

- Telegraph (1844, Samuel Morse)
  - Transmitted “What hath God wrought” between Washington, D.C. and Baltimore, Maryland.

Communication system breakthroughs

- Telephone (1875, Alexander Graham Bell)
  - Made real-time transmission of speech by electrical encoding and replication of sound

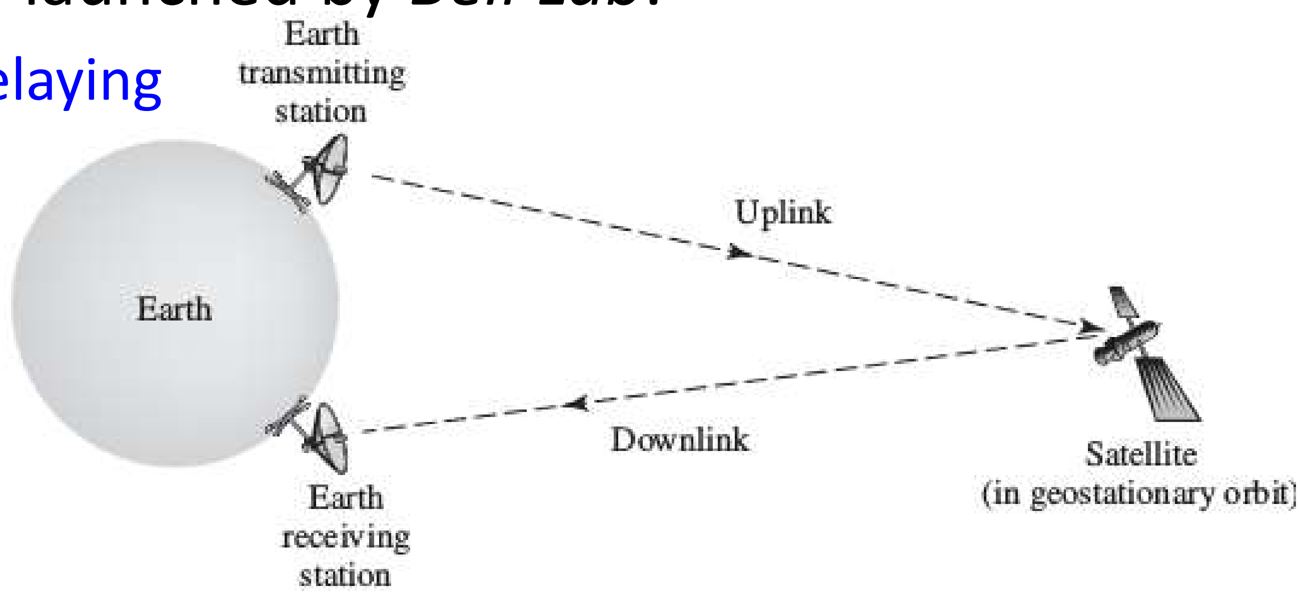


Communication system breakthroughs

- Radio (1901,Marconi) and TV (1925, Jenkins) broadcasting
- 1906:
  - AM radio
- 1935:
  - FM radio
- 1953
  - Color TV
- Recent innovation: internet radio/TV

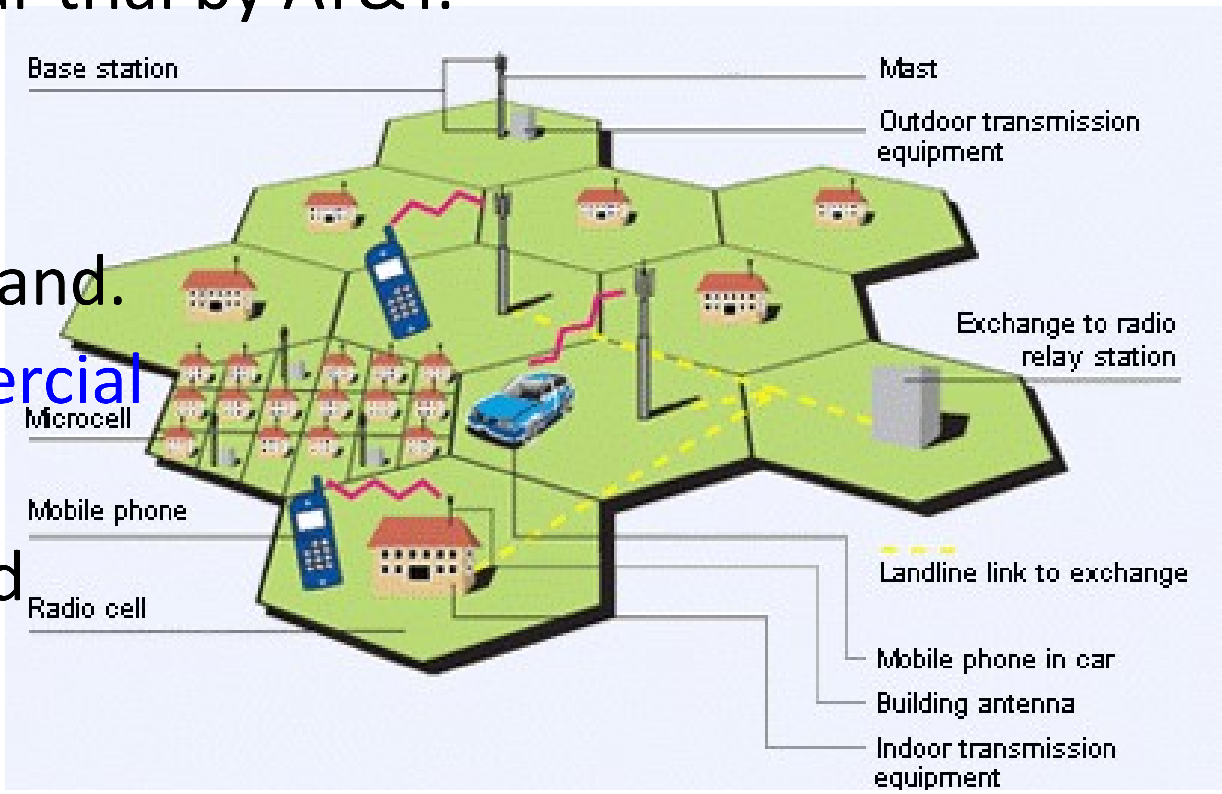
Communication system breakthroughs

- Satellite Communication
  - 1955: John R. *Pierce proposed*
  - 1957: *Sputnik I* launched by *Soviet U* transmitted telemetry signals for 21 days.
  - 1962: *Telstar I* launched by *Bell Lab*.
    - Capable of relaying TV programs across the Atlantic.



Communication system breakthroughs

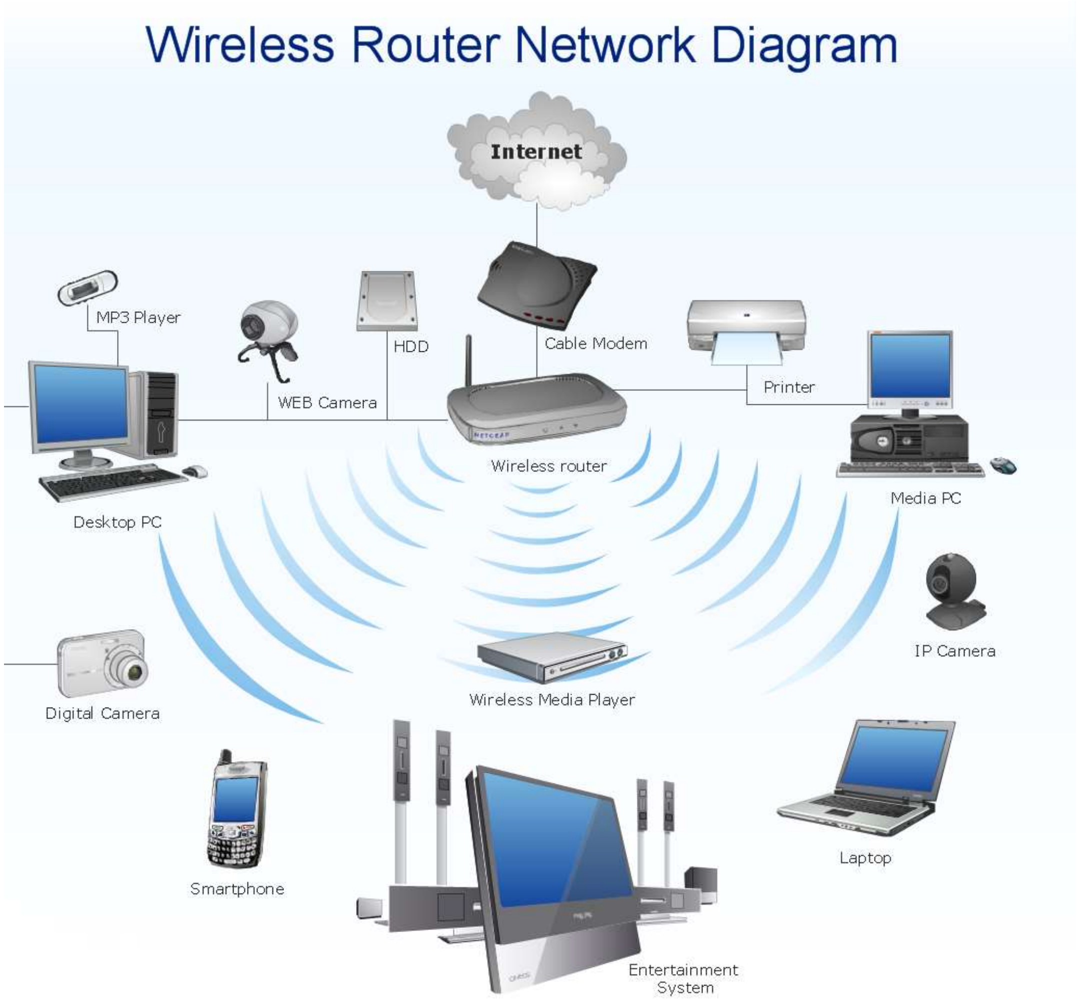
- Cellular Communication
  - 1947: cellular concept first proposed at Bell labs.
  - 1978: first cellular trial by AT&T.
  - 1991:first GSM cellular service launched in Finland.
  - 1996: 1st commercial CDMA cellular service launched in Finland.



Communication system breakthroughs

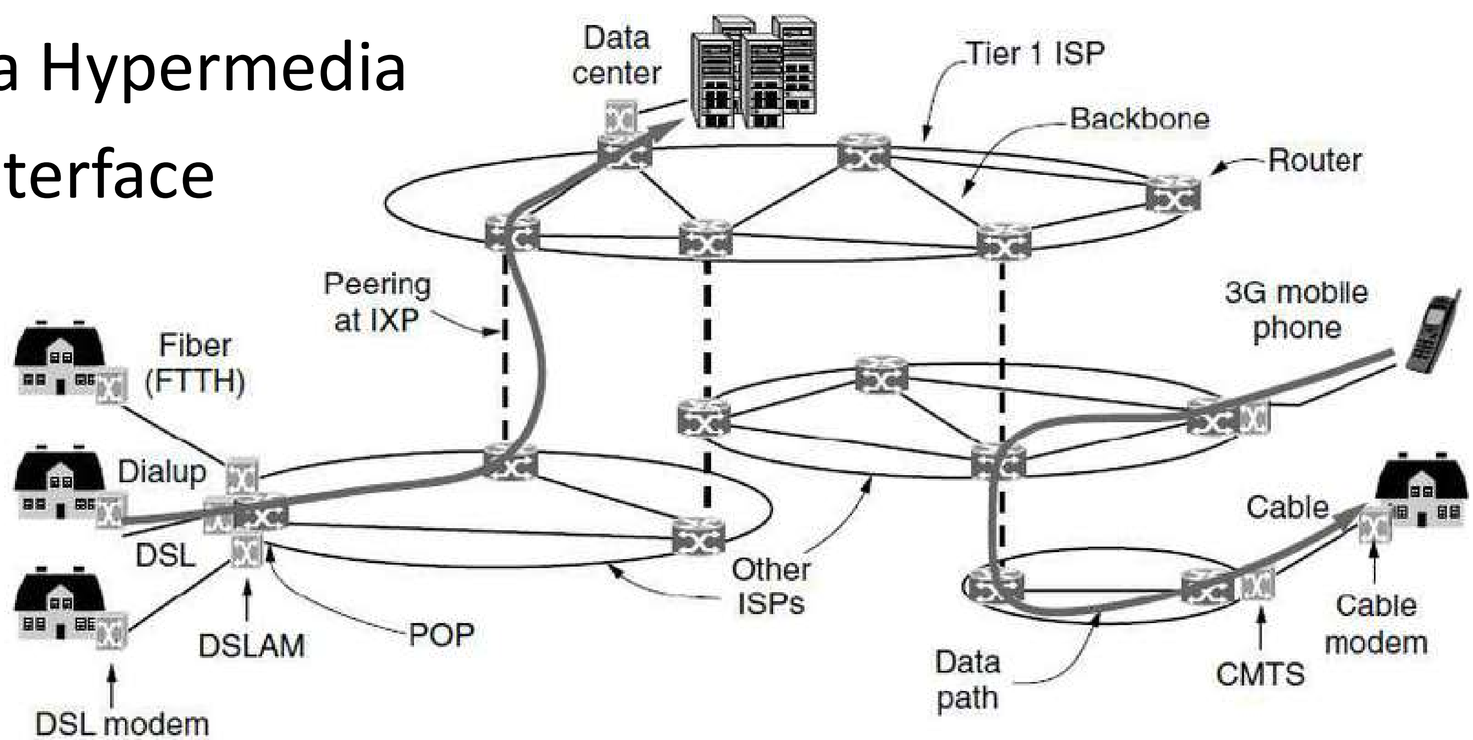
- Wireless LAN
  - 1971: 1st wireless computer network

    1997- to date: A No. of different IEEE Wireless LAN standards.

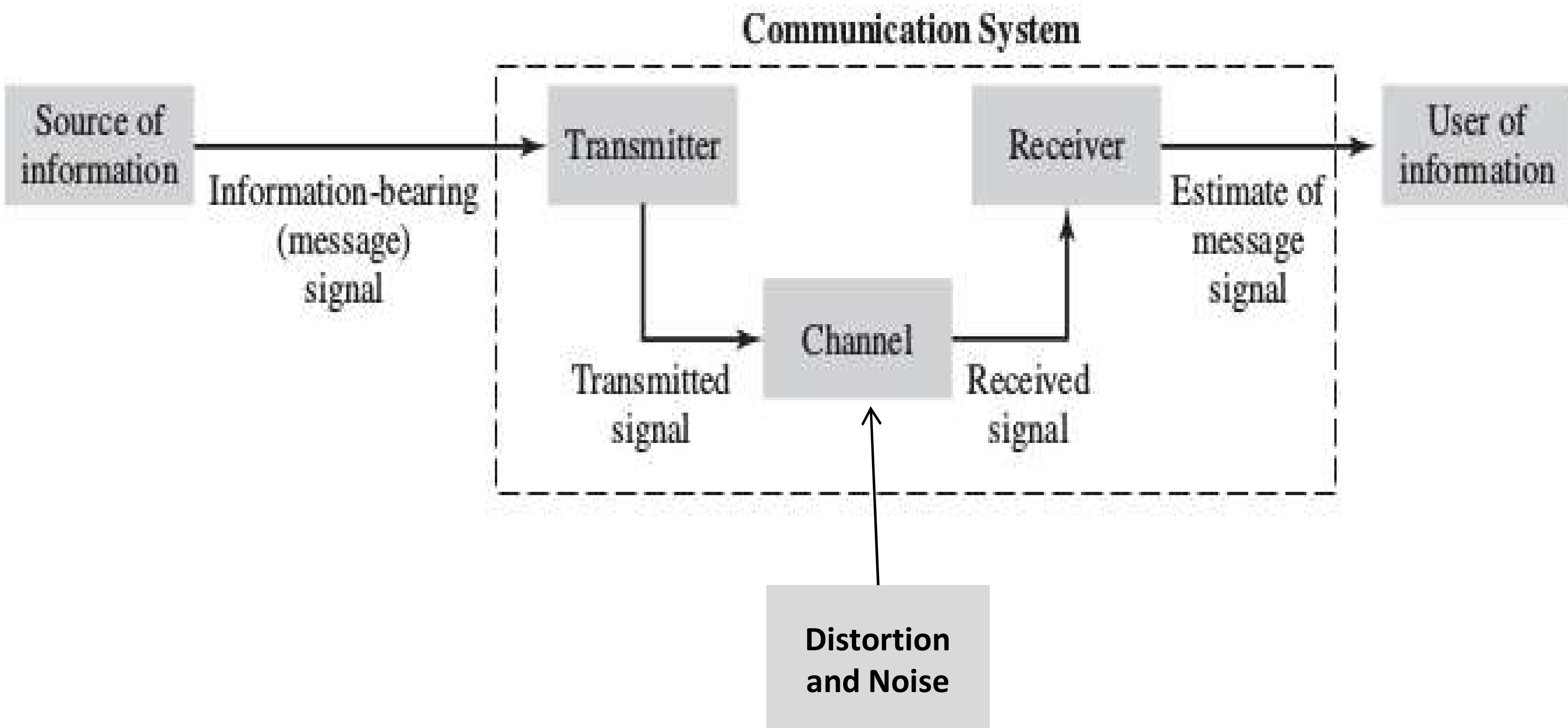


Communication system breakthroughs

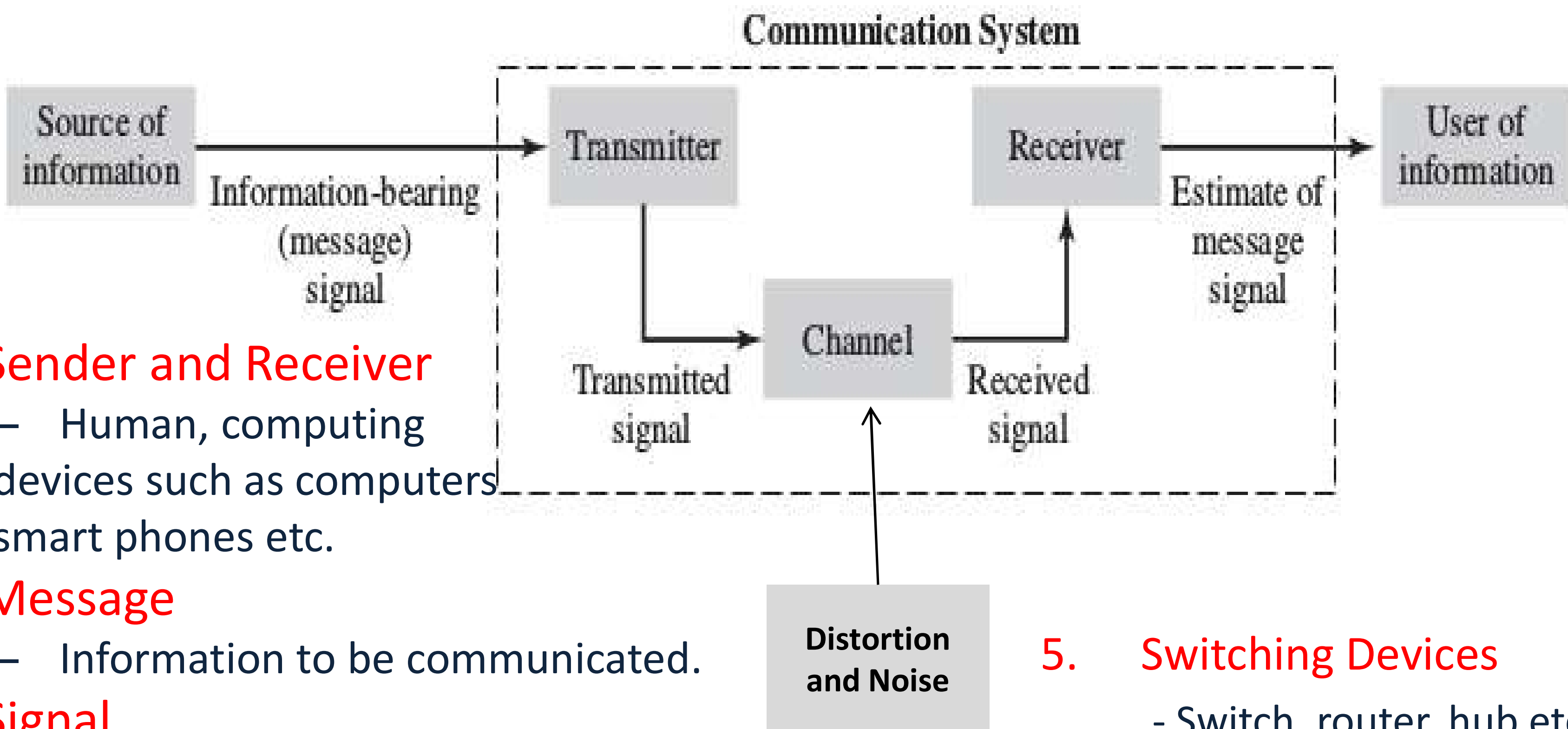
- Internet: Network of Networks
  - 1971: ARPANET put into service.
    - Later renamed the Internet, in 1985.
  - 1990: Tim Berners-Lee proposed a Hypermedia software interface to Internet WWW



Components of Communication systems

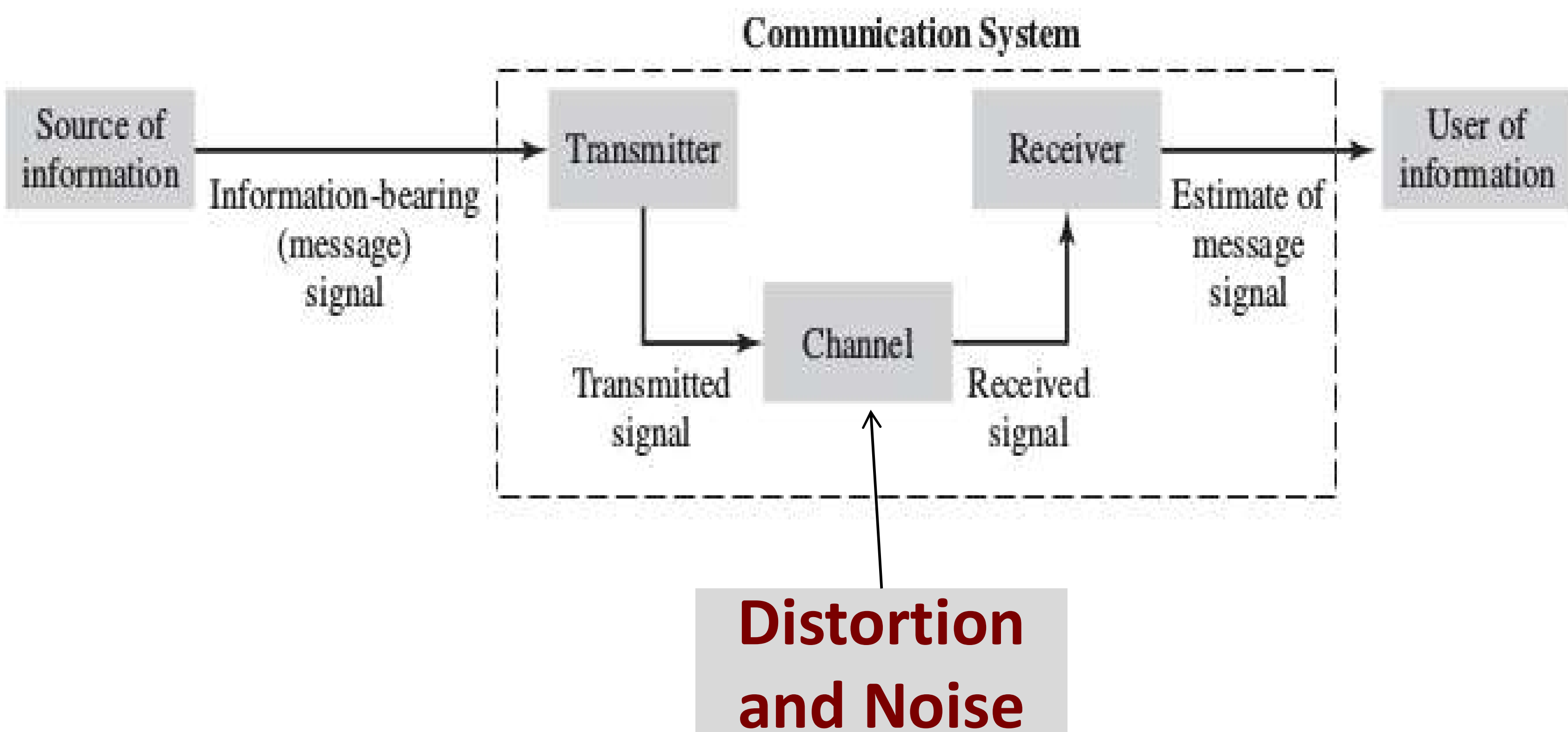


Components of Communication systems



1. Sender and Receiver
   – Human, computing devices such as computers smart phones etc.
2. Message
   – Information to be communicated.
3. Signal
   – Electrical, Optical, Sound
4. Medium/Channel
   – Air, Wire, Fiber, Vacuum
5. Switching Devices
   - Switch, router, hub etc.
6. Protocols: Rules of Communication
   – HTTP, SMTP, FTP etc.

Components of Communication systems



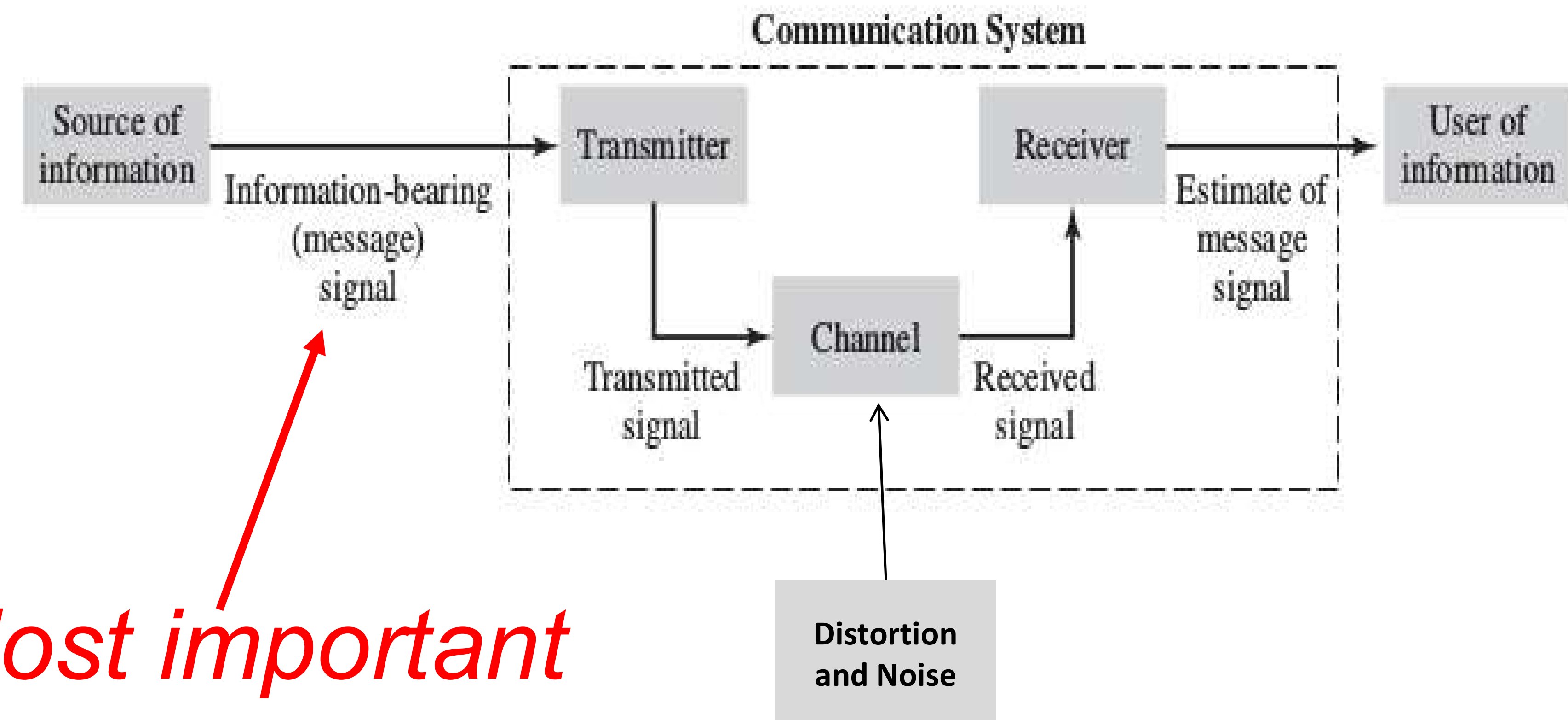
Main challenges

Challenges of Communication systems

1. Distortion
   - –systematic undesirable changes in signals
   - –Linear or non-linear

2. Noise
   - Unwanted signal that interfere with the transmitted signal
   - Random signals from internal or external sources

Components of Communication systems



Most important component

Messages/Signals: Definition

- A signal is a set of information or data.

- A signal is a function of independent variables that carry some information.

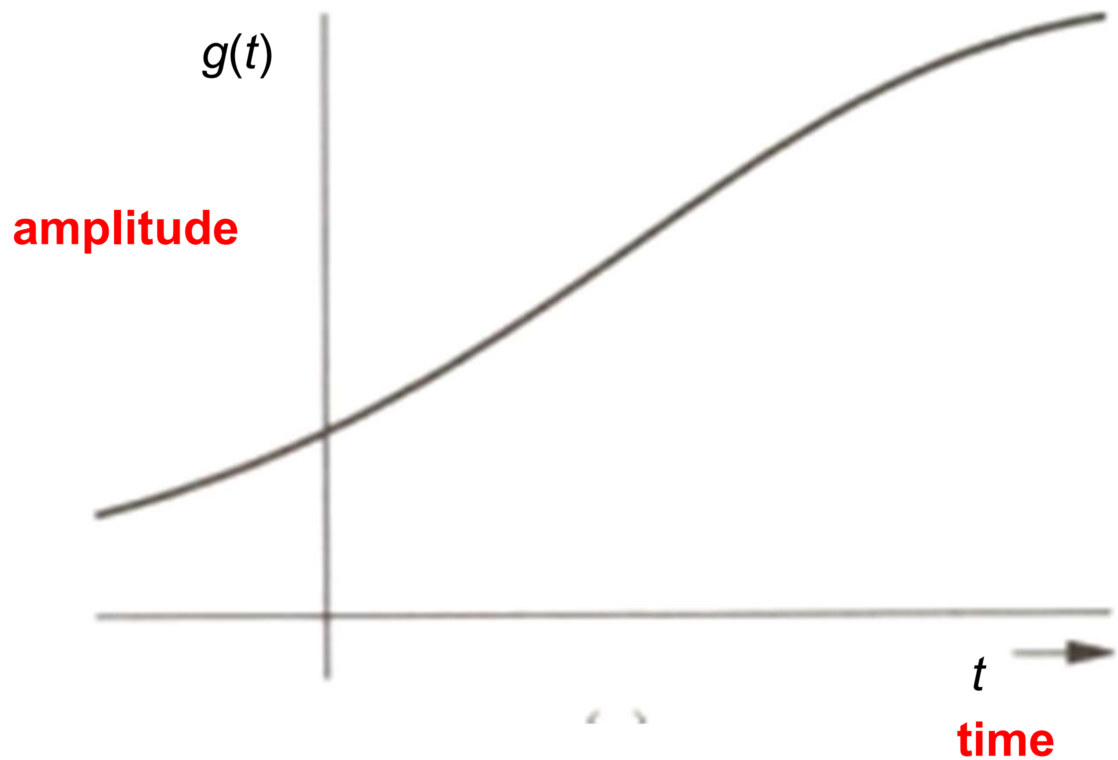
- A signal is a physical quantity that varies with time, space or any other independent variable by which information can be conveyed.

Example of Signals

- Voice signal
- Telephone or television signal
- Monthly sales figure
- Opening or closing stock prices
- Charge density over a surface

- In this course we deal with signals that are functions of time.

Signal representation: Time Domain



Classification of Signals

- Based on continuity in time axis
  - Continuous time
  - Discrete time
- Based on continuity in amplitude axis
  - Continuous amplitude
  - Discrete amplitude

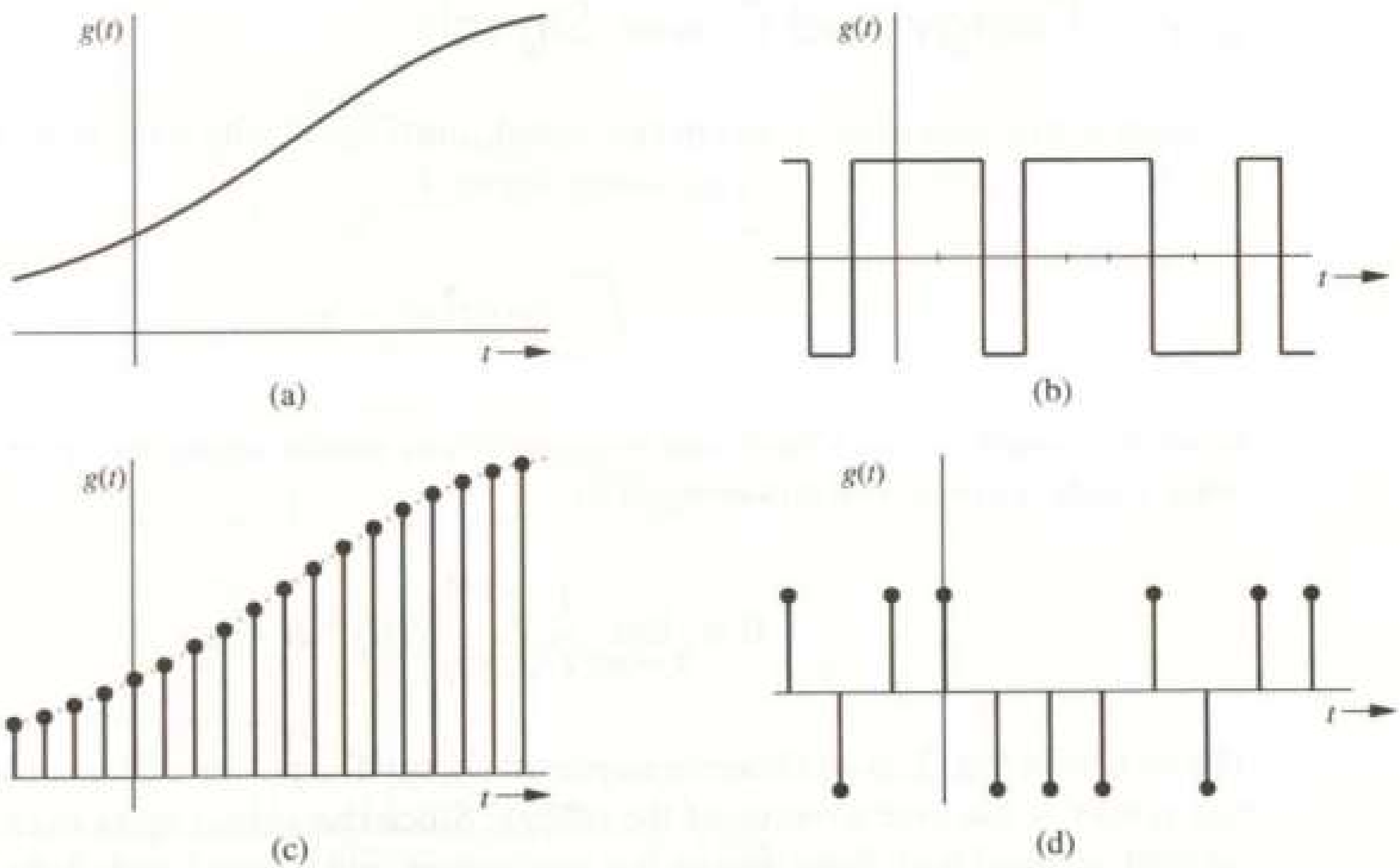
Classification of Signals

Continuous amplitude

Discrete amplitude

Continuous time

Discrete time



Analog and Digital Signal

Analog Signal

- Continuous amplitude, i.e., takes any value in a continuous range.
- May be both continuous and discrete time.

Digital Signal

- Discrete amplitude, i.e., amplitude can take only a finite number of values.
- Values need not be always integer.
- Not necessarily always binary, rather M-ary.
- May be both continuous and discrete time.

Analog and Digital Signal: Examples

	Analog	Digital
Thermometer		
Clock		
Blood Pressure Monitor